25900648 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCSTA200 - STATISTICAL DISTRIBUTIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **10** questions

Each question carries **3** marks

1. Given the following probability distribution: [A] / [CO1]

X	0	1	2	3	4	5	6	7
f(x)	0	c	2c	2c	3c	c^2	$2c^2$	$7c^2 + c$

Find the value of c.

2. What is geometric distribution? Write the expression for its mean and variance. [K] / [CO1]

3. Define a standardized Poisson random variable. [K] / [CO1]

4. Explain mathematical expectation of a continuous random variable. How do you calculate the mean of a continuous random variable? [U] / [CO1]

5. Explain central limit theorem. [A] / [CO1]

6. If X follows uniform distribution over (-3, 3), find P(|X| < 2). [U] / [CO1]

7. Define standard Laplace distribution and two parameter Laplace distribution. [K] / [CO1]

8. Find the characteristic function of the distribution with p.m.f $f(x) = pq^{x-1}, \quad x = 1, 2, 3, \ldots where \;\; p + q = 1.$ [A] / [CO2]

9. Explain the importance of standard error in statistics. [U] / [CO3]

10. Define Student's-t statistic and state its probability density function. [U] / [CO3]

11. If X is normally distributed with mean of X is 12 and S.D is 4. Find a and b such that $P[a < X < b] = 0.50$ and $P[X > b] = 0.25$. [A] / [CO1]

12. If 1 and 2 are modes of poisson distribution. Find $P(X = 0)$. [A] / [CO1]

[3x10 = 30]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **7** marks

13. What is the relation between c.d.f. and p.d.f. of a random variable? Find the c.d.f. of [A] / [CO1]

$$f(x) = \begin{cases} \frac{1}{\theta} & 0 \le x \le \theta \\ 0 & elsewhere \end{cases}$$

14. Define characteristic function. State and prove any three properties of characteristic function. [U] / [CO2]

15. State and prove lack of memory property of exponential distribution. [U] / [CO1]

16. Suppose that X is a r.v with p.d.f $f(x) = \begin{cases} 2\,x & 0 < x < 1 \\ 0 & elsewhere \end{cases}$. Find the p.d.f of (1) $Y = 3X + 1$ and (2)$Y = 8X^3$. [A] / [CO1]

17. State and prove the additive property of Chi-square distribution. [An] / [CO3]

18. A and B play a game in which their chance of winning are in the ratio 3 : 2 . Find the chance of winning A at least 3 games out of five games played. [A] / [CO1]

[7x4 = 28]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

19. (a) Define F-statistic. What is its p.d.f. ? Explain two important uses of it in staistical analysis. [A] / [CO3]

(b) Prove that if $n_1 = n_2$, the median of F-distribution is at F=1 and that the quartiles Q_1 and Q_3 satisfy the condition $Q_1 Q_3 = 1$.

20. Fit a normal distribution to the following data and obtain the expected frequencies : [A] / [CO4]

Class	60-65	65-70	70-75	75-80	80-85	85-90	90-95	95-100
Frequency	3	21	150	335	326	135	26	4

[12x1 = 12]